Use Case Flow
Online Quiz Management System

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UC-01 Register

Goal	Register a new student account
Actor	Student / User
Main flow	1. User opens the Online Quiz System. 2. Since the user is not already registered, the system routes them to "Register an Account". 3. User enters registration details and submits. 4. The system validates the input and creates the account. 5. The system redirects the user to "View Available Quizzes".
Alternate Flow	Already Registered At Step 2, if the user is already registered: 2a. The system routes the user to the Login flow instead (see UC-02 Login).

UC-02 Login

Goal	Log in to an existing account
Actor	Student / User, Admin
Main flow	1. User opens the Online Quiz System and, being already registered, selects "Login". 2. User enters their credentials. 3. The system authenticates the user (includes Authenticate User). 4. The system redirects the user to "View Available Quizzes" (or the admin dashboard, for Admin).
Alternate Flow	Invalid Credentials At Step 3, if authentication fails: 3a. The system displays an error message and keeps the user on the login screen to retry.

UC-03 View Available Quizzes & Select Category

Goal	Browse quizzes available to attempt
Actor	Student / User
Main flow	1. After registering or logging in, the user is shown the list of available quizzes. 2. User selects "Select Category / Subject" to narrow the list. 3. The system displays quizzes matching the selected category/subject.
Alternate Flow	No Quizzes in Category At Step 3, if no quizzes exist for the selected category: 3a. The system displays a message indicating no quizzes are available and prompts the user to choose another category.

UC-04 Attempt Quiz

Goal	Take a quiz and submit answers
Actor	Student / User
Main flow	1. User selects a quiz from the filtered list ("Select a Quiz") and selects "Start Quiz". 2. The system displays the multiple-choice questions along with a countdown timer (View Remaining Time). 3. User answers each question (includes Answer Questions); the system checks whether all questions have been answered. 4. Once all questions are answered, user selects "Submit Quiz" (includes Submit Quiz).
Alternate Flow	Not All Questions Answered At Step 3, if questions remain unanswered: 3a. The system returns the user to "Answer Multiple Choice Questions" until all are complete.
Alternate Flow	Time Expires At Step 2, if the countdown timer reaches zero before the user submits: 2a. The system automatically submits the quiz with the answers provided so far.

UC-05 View Quiz Results

Goal	View the automatically calculated quiz score
Actor	Student / User
Main flow	1. After the user submits a quiz, the system performs Automatic Evaluation / Score Calculation (includes Calculate Score and Store Results). 2. The system displays the quiz results (score) to the user.

UC-06 View Correct / Incorrect Answers

Goal	Review which answers were right or wrong
Actor	Student / User
Main flow	1. From the Display Quiz Results screen, user selects "View Correct / Incorrect Answers". 2. The system displays each question alongside the user's answer and the correct answer (extends Store Results).

UC-07 View Previous Attempts & Track Progress

Goal	Review quiz history and progress over time
Actor	Student / User
Main flow	1. From the Display Quiz Results screen, user selects "View Previous Attempts (History)" or "Track Progress". 2. The system retrieves the user's stored results and displays past attempts or a progress summary.
Alternate Flow	No Previous Attempts At Step 2, if the user has no quiz history: 2a. The system displays a message indicating no previous attempts exist.

UC-08 Logout

Goal	End the current session
Actor	Student / User, Admin
Main flow	1. User selects "Logout". 2. The system ends the session and returns the user to the start screen.

UC-09 Create Quiz

Goal	Create a new quiz
Actor	Admin
Main flow	1. Admin logs in and is authenticated (includes Authenticate User). 2. Admin selects "Create Quiz". 3. Admin defines the quiz details and adds or selects questions. 4. The system saves the quiz, making it available under "View Available Quizzes".
Alternate Flow	Authentication Fails At Step 1, if authentication fails: 1a. The system denies access to quiz creation and returns the admin to the login screen.

UC-10 Add / Edit / Delete Questions

Goal	Maintain the question bank
Actor	Admin
Main flow	1. Admin navigates to question management. 2. Admin selects "Add", "Edit", or "Delete" for a question. 3. The system validates the change and updates the question bank. 4. The system confirms the update.
Alternate Flow	Invalid Question Data At Step 3, if required fields are missing or invalid (e.g., no correct answer marked): 3a. The system displays an error and does not save the change.

UC-11 Manage Quiz Categories

Goal	Organize quizzes into categories/subjects
Actor	Admin
Main flow	1. Admin navigates to category management. 2. Admin adds, renames, or removes a category/subject. 3. The system updates the category list used in "Select Category / Subject".
Alternate Flow	Category In Use At Step 2, if Admin attempts to delete a category that has quizzes assigned to it: 2a. The system warns the Admin and requires reassignment or confirmation before deleting.

UC-12 Set Quiz Duration & Difficulty Level

Goal	Configure timing and difficulty for a quiz
Actor	Admin
Main flow	1. While creating or editing a quiz, Admin sets the duration (drives the countdown shown to users) and the difficulty level. 2. The system validates the values and saves them with the quiz.
Alternate Flow	Invalid Duration At Step 2, if the duration is zero, negative, or otherwise invalid: 2a. The system rejects the value and prompts the Admin to correct it.

UC-13 View User Results & Scores

Goal	Review all users' quiz results
Actor	Admin
Main flow	1. Admin navigates to "View User Results & Scores". 2. The system retrieves stored results for all users (includes Store Results). 3. The system displays user results and scores.
Alternate Flow	No Results Yet At Step 3, if no attempts have been recorded for a quiz: 3a. The system displays a message indicating no attempts exist yet.